SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

Use-Case Flow Document

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PROBLEM STATEMENT #61

Sustainability & Green Tech

Household Carbon Footprint & Target Tracker

Use-Case Flow Specification

Use Case:	Log Monthly Utility Data (UC-01)
Actor(s):	Resident (primary); system-included: Calculate CO2e Emissions
Preconditions:	1) The Resident is authenticated and logged in. 2) Utility data for the current reporting month has not already been submitted.
Postconditions:	The monthly log and its computed CO2e value are stored; the Resident's target-progress percentage is updated; a confirmation with the results is shown.

Main Success Scenario

1. Resident selects “Log Monthly Data” from the dashboard.

2. System displays a form requesting electricity (kWh), gas (therms), and transport (km) for the current month.

3. Resident enters the electricity, gas, and transport values and submits the form.

4. System validates that all entered values are non-negative numbers.

5. System invokes the included use case «include» Calculate CO2e Emissions, applying EPA conversion factors to the entered data.

6. System stores the monthly log together with the computed CO2e total.

7. System recalculates and updates the Resident's progress toward their annual carbon-reduction target.

8. System displays a confirmation showing the month's total CO2e and updated target progress.

9. Use case ends successfully.

Alternate Flow — 4a. Invalid Input

4a1. System detects a negative or non-numeric value in one or more fields and highlights the invalid field(s) with an error message.

4a2. Resident corrects the value(s) and resubmits the form.

4a3. Flow resumes at Main Success Scenario step 4 (validation).